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Module 5 Introduction

Lesson 5: Kernel Smoothers

Lesson 2: Local Regression

Reading: Local Regression Readings 10%

Video: Local Regression 6 min

Practice Assignment: Local Regression Quiz Submitted

Lab: Coding Exercise 10%

Module 5 Summative Assessment

Module 5 Summary

Your grade: 100%

Your best: 100% - Your highest score

Local Regression Quiz

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Instructions

Review Learning Objectives

1. What is a key characteristic of local regression?

- ☒ It fits simple models to localized subsets of the data.
- ☐ It requires a fixed number of data points for each data point.
- ☐ It uses the same coefficients for each data point.
- ☐ It always fits a global model to the entire dataset.

Correct

Correct: Local regression fits simple models to small parts of the data.

Submitted

Attempts

Try again

2. How does local regression handle non-linear patterns in the data?

- ☐ By transforming all features into linearly correlated variables.
- ☒ By fitting simple models within small, localized regions of the data, allowing for flexibility in the overall fit.
- ☐ By increasing the degree of the polynomial used to fit the data.
- ☐ By removing any non-linear features from the dataset.

Correct

Correct: Local regression allows non-linear patterns to be captured by fitting simple models over localized areas.

3. What role does the bandwidth play in local regression?

- ☐ It defines the degree of the polynomial used in the regression.
- ☐ It sets the number of local models to be fitted.
- ☐ It adjusts the weight given to each feature in the model.
- ☒ It determines the size of the neighborhood around each point to be used in the local model.

Correct

Correct: Bandwidth controls the size of the local area around each point for the regression.

1 / 1 point

View submission

1 / 1 point